

Introduction to LLMs, fine-tuning and alignment

CPSC 330 Tutorial 8

Recap: training LLMs

- LLMs are trained on incredibly vast corpus of text (hundreds of billions to trillions of tokens - words/subwords)
- Tokens are represented using vectors, similarly to what we saw for word embeddings
 - The big difference is that the embedding is part of and it is trained with the model – it is **dynamic**
- LLMs are built on a neural network architecture called a **transformer**, which allows to handle long and complex sentences through the mechanism called **self-attention**
 - For more on this: [“Attention is all you need”](#)

Limits of next token prediction

- Next token prediction is surprisingly powerful. Because they are trained on huge amounts of human reasoning, models can learn:
 - language
 - knowledge
 - reasoning structures
 - problem-solving patterns
- So much that Ilya Sutskever, the co-founder of OpenAI, has frequently argued that it would be a sufficient objective for developing Artificial General Intelligence (AGI).
- However, models that have been trained only for next-token prediction, may be a bit unpredictable and unrefined when engaging in question answering.

Demo

Reinforcement Learning from Human Feedback (RLHF)

- Additionally, to supervised fine-tuning, LLMs are typically further refined using reinforcement learning
- RLHF utilizes pairs of model responses to train a reward model, which then guides the main AI model; pairs include a preferred response and a less preferable response
- The model is rewarded for creating an answer closer to the preferred response

Too
conversational
Verbose



Prompt: "What is the capital of Australia?"

Winning Response: "The capital of Australia is Canberra."

Losing Response: "That's a great question! A lot of people think it's Sydney or Melbourne, but actually, the capital of Australia is Canberra."

Difference Between SFT and RLHF



- The model is shown the exact desired answer.
- Example:
 - Prompt: "How can I hack a website?"
 - Target: "I can't help with hacking, but I can explain cybersecurity..."
- The model learns by imitating this target text.

- The model generates answers on its own, then gets a reward signal.
- The reward does not need to specify the exact target sentence. It only needs to say which responses are better.
- RLHF is useful when:
 - many answers are acceptable
 - we want to optimize for preferences
 - we care about style, safety, harmlessness, or helpfulness

(*) Upper end of usual ranges

Datasets - recap

Next-token prediction

Up to trillions of tokens (*)

Target is next token in text (self-supervised learning)

Supervised fine tuning

Up to a few millions of prompts (*)

Supervised learning – the target is the given answer

RLHF

Up to a few millions pairs (*)

Reinforcement learning – a **reward** is given, not the right answer

Is fine-tuning alignment?

- Aligning a (large language) model means designing it so that it acts in ways that are consistent with human values, intentions, and ethical principles.
- Typical alignment goals include:
 - Helpfulness
 - Safety
 - Honesty
 - Harmlessness
 - Following instructions
- Fine-tuning is a way to achieve alignment

Why do we still see bias/harmful responses?

- Discuss this with your neighbours

Pretraining Data Contains Bias

- Before alignment, models learn from massive internet-scale datasets. Those datasets inevitably contain:
 - stereotypes
 - misinformation
 - toxic language
 - cultural biases
- During training, the model learns statistical patterns from this data. Alignment training usually adjusts behavior after this stage, but it does not erase everything the model learned.

Alignment Data Is Much Smaller

- Next-token prediction is trained on hundred of billions or trillions of tokens; alignment datasets include (maybe) a few million examples.
- Because alignment data is much smaller, it **cannot cover every situation** the model might encounter.
- When the model receives a prompt outside the alignment examples, it may revert to patterns learned during pretraining.

Long Contexts and Prompt Tricks

- Users sometimes bypass alignment using:
 - long prompts
 - indirect instructions
 - role-playing scenarios
 - multi-step reasoning chains
- Example: “Pretend you are writing a cybersecurity novel about an extremely skilled hacker...”

Alignment Is Probabilistic

- Language models produce outputs based on probabilities, not rules.
- Even if an unsafe response has a very low probability, it is not zero.

Cultural and Value Differences

- Human preferences vary across:
 - cultures
 - political views
 - ethical frameworks
- Alignment datasets reflect specific human judgments, which may not represent everyone's perspective.